

Mathematics Notes: Fourier Series & Periodic Functions

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1 Introduction to Fourier Series (Page 1)

Definition of Periodic Function

A function $f(x)$ is said to be periodic of period $P > 0$ if:

$$f(x + P) = f(x) \quad \text{for all } x$$

Example 1: Determine the Period of $\cos x$

To find P such that $\cos(x + P) = \cos x$:

$$\cos(x + P) - \cos x = 0$$

Using the trigonometric identity $\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$:

$$\begin{aligned} -2 \sin\left(\frac{2x + P}{2}\right) \sin\left(\frac{P}{2}\right) &= 0 \\ \implies \sin\left(\frac{P}{2}\right) &= 0 \implies \frac{P}{2} = n\pi \implies P = 2n\pi \end{aligned}$$

Thus, 2π is the fundamental (smallest positive) period.

Theorem 1: Linear Combination of Periodic Functions

If $f(x)$ and $g(x)$ both have period P , then show that $h(x) = af(x) + bg(x)$ (where a, b are constants) also has period P .

Proof: Given $f(x + P) = f(x)$ and $g(x + P) = g(x)$:

$$\begin{aligned} h(x + P) &= af(x + P) + bg(x + P) \\ &= af(x) + bg(x) \\ &= h(x) \end{aligned}$$

Hence, $h(x)$ has period P .

2 Properties of Periodic Functions (Page 2)

Theorem 2: Scaling the Argument

If $f(x)$ has period P , then:

1. $f(ax)$ has period $\frac{P}{a}$
2. $f\left(\frac{x}{b}\right)$ has period bP

Proof for $f(ax)$: Let $g(x) = f(ax)$. Then:

$$g\left(x + \frac{P}{a}\right) = f\left(a\left(x + \frac{P}{a}\right)\right) = f(ax + P) = f(ax) = g(x)$$

Thus, $g(x)$ has period $\frac{P}{a}$.

Proof for $f\left(\frac{x}{b}\right)$: Let $h(x) = f\left(\frac{x}{b}\right)$. Then:

$$h(x + bP) = f\left(\frac{x + bP}{b}\right) = f\left(\frac{x}{b} + P\right) = f\left(\frac{x}{b}\right) = h(x)$$

Thus, $h(x)$ has period bP .

3 Orthogonality Relations of Sine and Cosine (Page 3)

For integers n, m :

Integrals over a full period $[\alpha, \alpha + 2\pi]$

1. $\int_{\alpha}^{\alpha+2\pi} \sin(nx) dx = 0$
2. $\int_{\alpha}^{\alpha+2\pi} \cos(nx) dx = 0$ (for $n \neq 0$)
3. $\int_{\alpha}^{\alpha+2\pi} \cos(mx) \cos(nx) dx = 0$ (for $n \neq m$)
4. $\int_{\alpha}^{\alpha+2\pi} \sin(mx) \sin(nx) dx = 0$ (for $n \neq m$)
5. $\int_{\alpha}^{\alpha+2\pi} \sin(mx) \cos(nx) dx = 0$ (for all n, m)

Square Integrals

1. $\int_{\alpha}^{\alpha+2\pi} \cos^2(nx) dx = \pi$ ($n \neq 0$)
 2. $\int_{\alpha}^{\alpha+2\pi} \sin^2(nx) dx = \pi$ ($n \neq 0$)
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Fourier Series Definition

The Fourier series of a periodic function $f(x)$ with period 2π is:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

4 Euler's Formulas for Fourier Coefficients (Page 4)

1. Derivation of a_0

Integrating both sides of the Fourier series over $[\alpha, \alpha + 2\pi]$:

$$\int_{\alpha}^{\alpha+2\pi} f(x) dx = \int_{\alpha}^{\alpha+2\pi} a_0 dx + \sum_{n=1}^{\infty} \left(a_n \int_{\alpha}^{\alpha+2\pi} \cos(nx) dx + b_n \int_{\alpha}^{\alpha+2\pi} \sin(nx) dx \right)$$

Using orthogonality, the sum terms vanish:

$$\int_{\alpha}^{\alpha+2\pi} f(x) dx = a_0(2\pi) + 0 + 0 \implies a_0 = \frac{1}{2\pi} \int_{\alpha}^{\alpha+2\pi} f(x) dx$$

2. Derivation of a_n

Multiplying by $\cos(mx)$ and integrating:

$$\int_{\alpha}^{\alpha+2\pi} f(x) \cos(mx) dx = a_n \cdot \pi \implies a_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cos(nx) dx$$

3. Derivation of b_n

Multiplying by $\sin(mx)$ and integrating:

$$\int_{\alpha}^{\alpha+2\pi} f(x) \sin(mx) dx = b_n \cdot \pi \implies b_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \sin(nx) dx$$

5 Fourier Series Examples (Pages 5 & 6)

Example 1: Square Wave

Find the Fourier series of:

$$f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$$

Coefficients:

$$\begin{aligned}a_0 &= \frac{1}{2\pi} \left[\int_{-\pi}^0 (-k) dx + \int_0^{\pi} k dx \right] = \frac{1}{2\pi} \left[-k(0 - (-\pi)) + k(\pi - 0) \right] = 0 \\a_n &= \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \cos(nx) dx + \int_0^{\pi} k \cos(nx) dx \right] = 0 \\b_n &= \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \sin(nx) dx + \int_0^{\pi} k \sin(nx) dx \right] \\&= \frac{k}{\pi} \left[\left[\frac{\cos(nx)}{n} \right]_{-\pi}^0 - \left[\frac{\cos(nx)}{n} \right]_0^{\pi} \right] \\&= \frac{k}{n\pi} \left[(1 - \cos(-n\pi)) - (\cos(n\pi) - 1) \right] \\&= \frac{2k}{n\pi} (1 - \cos(n\pi)) = \frac{2k}{n\pi} (1 - (-1)^n)\end{aligned}$$

Fourier Series Expansion:

$$f(x) = \frac{2k}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \sin(nx)$$

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Example 2: Piecewise Constant Function

Find the Fourier series for $f(x)$ with period $T = 4$ ($\alpha = -2$):

$$f(x) = \begin{cases} 0, & -2 < x < -1 \\ k, & -1 < x < 1 \\ 0, & 1 < x < 2 \end{cases}$$

Coefficients:

$$\begin{aligned}a_0 &= \frac{1}{4} \int_{-1}^1 k dx = \frac{k}{4}(2) = \frac{k}{2} \\a_n &= \frac{1}{2} \int_{-1}^1 k \cos\left(\frac{n\pi x}{2}\right) dx = \frac{k}{2} \left[\frac{2}{n\pi} \sin\left(\frac{n\pi x}{2}\right) \right]_{-1}^1 = \frac{2k}{n\pi} \sin\left(\frac{n\pi}{2}\right) \\b_n &= \frac{1}{2} \int_{-1}^1 k \sin\left(\frac{n\pi x}{2}\right) dx = 0 \quad (\text{due to odd symmetry})\end{aligned}$$

Fourier Series:

$$f(x) = \frac{k}{2} + \frac{2k}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{n\pi x}{2}\right)$$

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6 Fourier Series for Arbitrary Period T (Pages 7 & 8)

Let $f(x)$ be a periodic function of period T on the interval $[\alpha, \alpha + T]$.

By change of variable $v = \frac{2\pi x}{T} \implies x = \frac{vT}{2\pi}$:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{2n\pi x}{T}\right) + b_n \sin\left(\frac{2n\pi x}{T}\right) \right)$$

where the general Euler coefficients are:

$$\begin{aligned} a_0 &= \frac{1}{T} \int_{\alpha}^{\alpha+T} f(x) dx \\ a_n &= \frac{2}{T} \int_{\alpha}^{\alpha+T} f(x) \cos\left(\frac{2n\pi x}{T}\right) dx \\ b_n &= \frac{2}{T} \int_{\alpha}^{\alpha+T} f(x) \sin\left(\frac{2n\pi x}{T}\right) dx \end{aligned}$$

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7 Even and Odd Functions & Half-Range Expansions (Pages 9,10)

Definitions

- **Even Function:** $f(-x) = f(x)$ (Symmetric about the y -axis).
- **Odd Function:** $f(-x) = -f(x)$ (Symmetric about the origin).

Properties of Integrals

- If $f(x)$ is even: $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$
- If $f(x)$ is odd: $\int_{-a}^a f(x) dx = 0$

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Fourier Series for Even Functions

If $f(x)$ is an **even** function:

$$\begin{aligned} a_0 &= \frac{2}{T} \int_0^{T/2} f(x) dx \\ a_n &= \frac{4}{T} \int_0^{T/2} f(x) \cos\left(\frac{2n\pi x}{T}\right) dx \\ b_n &= 0 \end{aligned}$$

The series contains **only cosine terms**.

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Fourier Series for Odd Functions

If $f(x)$ is an **odd** function:

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{4}{T} \int_0^{T/2} f(x) \sin\left(\frac{2n\pi x}{T}\right) dx$$

The series contains **only sine terms**.

8 Transition from Fourier Series to Fourier Integral (Page 1)

For a function $f(x)$ defined on $(-L, L)$, the Fourier series expansion is:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

where the coefficients are given by:

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Substituting a_0, a_n , and b_n back into $f(x)$:

$$f(x) = \frac{1}{2L} \int_{-L}^L f(v) dv + \frac{1}{L} \sum_{n=1}^{\infty} \left[\int_{-L}^L f(v) \cos\left(\frac{n\pi v}{L}\right) dv \right] \cos\left(\frac{n\pi x}{L}\right) + \frac{1}{L} \sum_{n=1}^{\infty} \left[\int_{-L}^L f(v) \sin\left(\frac{n\pi v}{L}\right) dv \right] \sin\left(\frac{n\pi x}{L}\right)$$

Let $\omega_n = \frac{n\pi}{L}$, then $\Delta\omega_n = \omega_n - \omega_{n-1} = \frac{\pi}{L} \implies \frac{1}{L} = \frac{\Delta\omega_n}{\pi}$.

As $L \rightarrow \infty$, $\Delta\omega \rightarrow 0$, converting the summation into an integral:

$$f(x) = \frac{\Delta\omega}{2\pi} \int_{-L}^L f(v) dv + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\int_{-L}^L f(v) \cos(\omega_n v) dv \right) \cos(\omega_n x) \Delta\omega + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\int_{-L}^L f(v) \sin(\omega_n v) dv \right) \sin(\omega_n x) \Delta\omega$$

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9 Fourier Cosine and Sine Integral Representations (Page 2)

In limit as $L \rightarrow \infty$:

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \left[\int_0^{\infty} f(v) \cos(\omega v) dv \right] \cos(\omega x) d\omega + \frac{2}{\pi} \int_0^{\infty} \left[\int_0^{\infty} f(v) \sin(\omega v) dv \right] \sin(\omega x) d\omega$$

Fourier Cosine Transform Pair

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty f_c(\omega) \cos(\omega x) d\omega \quad (\text{Inverse Fourier Cosine Transform})$$

$$f_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos(\omega x) dx \quad (\text{Fourier Cosine Transform})$$

Fourier Sine Transform Pair

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty f_s(\omega) \sin(\omega x) d\omega \quad (\text{Inverse Fourier Sine Transform})$$

$$f_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin(\omega x) dx \quad (\text{Fourier Sine Transform})$$

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10 Examples of Fourier Transforms (Page 3)

Example 1: Rectangular Pulse

Find the Fourier Cosine Transform of:

$$f(x) = \begin{cases} k, & 0 < x < a \\ 0, & x > a \end{cases}$$

Solution:

$$f_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^a k \cos(\omega x) dx = \sqrt{\frac{2}{\pi}} k \left[\frac{\sin(\omega x)}{\omega} \right]_0^a = k \sqrt{\frac{2}{\pi}} \frac{\sin(\omega a)}{\omega}$$

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Example 2: Exponential Decay

Find the Fourier Cosine Transform of $f(x) = e^{-\omega x}$ (or e^{-ax} for $a > 0$):

$$f_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x} \cos(\omega x) dx$$

Using the standard integral identity $\int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$:

$$f_c(\omega) = \sqrt{\frac{2}{\pi}} \left[\frac{e^{-x}}{1 + \omega^2} (-\cos(\omega x) + \omega \sin(\omega x)) \right]_0^\infty = \sqrt{\frac{2}{\pi}} \left(0 + \frac{1}{1 + \omega^2} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{1}{1 + \omega^2} \right)$$

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11 Fourier Transform of Derivatives: Cosine & Sine (Page 4)

Theorem

Let $f(x)$ be continuous and absolutely integrable on $[0, \infty)$, and $f(x) \rightarrow 0$ as $x \rightarrow \infty$.

1. $\mathcal{F}_c\{f'(x)\} = \omega \mathcal{F}_s\{f(x)\} - \sqrt{\frac{2}{\pi}}f(0)$
2. $\mathcal{F}_s\{f'(x)\} = -\omega \mathcal{F}_c\{f(x)\}$

Proof for $\mathcal{F}_c\{f'(x)\}$:

$$\begin{aligned}\mathcal{F}_c\{f'(x)\} &= \sqrt{\frac{2}{\pi}} \int_0^\infty f'(x) \cos(\omega x) dx \\ &= \sqrt{\frac{2}{\pi}} \left[\left[\cos(\omega x) f(x) \right]_0^\infty + \omega \int_0^\infty f(x) \sin(\omega x) dx \right] \\ &= \sqrt{\frac{2}{\pi}} (0 - f(0)) + \omega \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin(\omega x) dx \\ &= \omega \mathcal{F}_s\{f(x)\} - \sqrt{\frac{2}{\pi}} f(0)\end{aligned}$$

12 Complex Form of Fourier Integral & Transform (Page 5)

The Fourier Integral representation can be written in exponential form:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^\infty \int_{-\infty}^\infty f(v) e^{i(x-v)\omega} dv d\omega$$

Splitting into transform pairs:

$$\begin{aligned}\hat{f}(\omega) &= \mathcal{F}\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty f(v) e^{-i\omega v} dv \\ f(x) &= \mathcal{F}^{-1}\{\hat{f}(\omega)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \hat{f}(\omega) e^{i\omega x} d\omega\end{aligned}$$

13 Complex Fourier Transform Example (Page 6)

Example: Double-Sided Exponential

Find the Fourier transform of:

$$f(x) = \begin{cases} e^x, & x < 0 \\ e^{-x}, & x \geq 0 \end{cases} \quad (\text{i.e., } f(x) = e^{-|x|})$$

Solution:

$$\begin{aligned}
\hat{f}(\omega) &= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^0 e^x e^{-i\omega x} dx + \int_0^{\infty} e^{-x} e^{-i\omega x} dx \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^0 e^{(1-i\omega)x} dx + \int_0^{\infty} e^{-(1+i\omega)x} dx \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[\left[\frac{e^{(1-i\omega)x}}{1-i\omega} \right]_{-\infty}^0 + \left[\frac{e^{-(1+i\omega)x}}{-(1+i\omega)} \right]_0^{\infty} \right] \\
&= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{1-i\omega} - 0 + 0 - \left(-\frac{1}{1+i\omega} \right) \right] \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{1-i\omega} + \frac{1}{1+i\omega} \right) \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{2}{1+\omega^2} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{1}{1+\omega^2} \right)
\end{aligned}$$

14 Fourier Transform of Derivative & Convolution Theorem (Pages 7 & 8)

Derivative Property of Complex Fourier Transform

Let $f(x)$ be continuous on the real line and $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Then:

$$\mathcal{F}\{f'(x)\} = i\omega \mathcal{F}\{f(x)\} = i\omega \hat{f}(\omega)$$

Proof:

$$\begin{aligned}
\mathcal{F}\{f'(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(x) e^{-i\omega x} dx \\
&= \frac{1}{\sqrt{2\pi}} \left[\left[f(x) e^{-i\omega x} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x) (-i\omega) e^{-i\omega x} dx \right] \\
&= \frac{1}{\sqrt{2\pi}} (0 - 0) + i\omega \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\
&= i\omega \hat{f}(\omega)
\end{aligned}$$

Convolution Theorem

Definition of Convolution: The convolution of two functions $f(x)$ and $g(x)$, denoted $(f * g)(x)$, is defined as:

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-p)g(p) dp$$

Theorem Statement:

$$\mathcal{F}\{(f * g)(x)\} = \sqrt{2\pi} \hat{f}(\omega) \hat{g}(\omega)$$

Proof:

$$\begin{aligned} \mathcal{F}\{(f * g)(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(p)g(x-p) dp \right] e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)g(x-p)e^{-i\omega x} dx dp \end{aligned}$$

Substitute $q = x - p \implies x = q + p$ and $dx = dq$:

$$\begin{aligned} \mathcal{F}\{(f * g)(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)g(q)e^{-i\omega(p+q)} dp dq \\ &= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} f(p)e^{-i\omega p} dp \right) \left(\int_{-\infty}^{\infty} g(q)e^{-i\omega q} dq \right) \\ &= \sqrt{2\pi} \hat{f}(\omega) \hat{g}(\omega) \end{aligned}$$